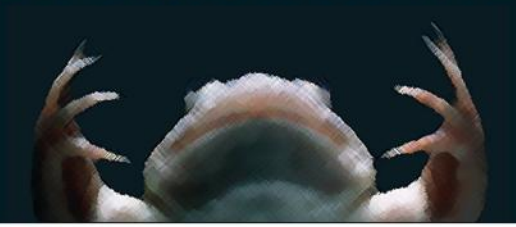
COLOUR VISION IN THE DARK: RETINAL COMPUTATIONS UNDERLYING ROD-DRIVEN CHROMATIC PROCESSING



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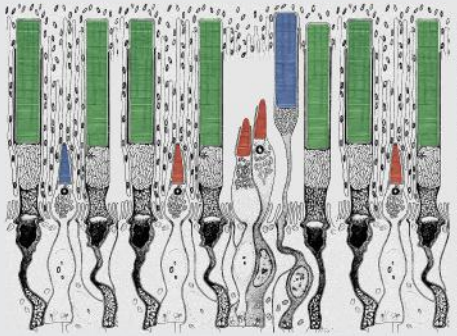
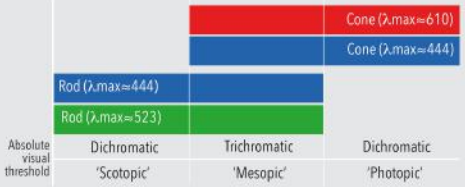


Adapted from Nilsson 1964 J Ultrastruct Res 10(5-6): 390-416

Frogs and some salamanders are unique among vertebrates in having two rod types with different spectral sensitivities.

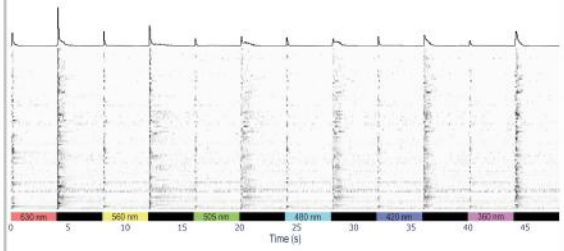


This retinal configuration allows frogs to discriminate colours at scotopic light levels in some visually guided behaviours (Yovanovich et al 2017 *Phil Trans R Soc B*) but the underlying circuitry remains unknown, so we are attempting to uncover the computations and connections involved to find out:

How is wavelength encoded at the retinal output under different illumination regimes?

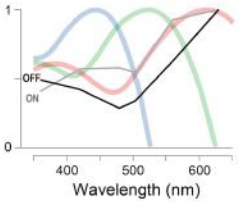
We are using 64x64 electrode arrays to record responses to light flashes of different intensities in ganglion cells of dark-adapted retinas of *Xenopus laevis*. We use a 6-LED stimulator to deliver full field colour flashes and colour noise, with addition of neutral density filters in the optical path to decrease light intensity.

'PHOTOPIC'

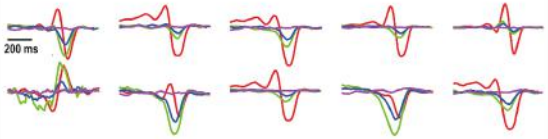


The response pattern at relatively high illumination is driven by a majority of ON-OFF units, which are most usually transient ON / sustained OFF.

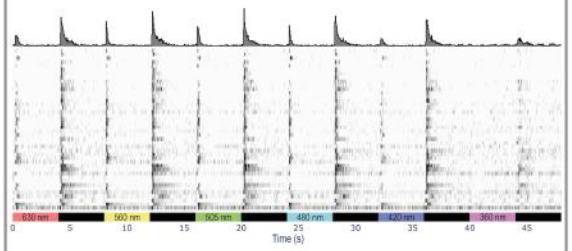
The stronger response of the overall cell population at longer wavelengths is consistent with red-sensitive being the most abundant cone spectral type.



Colour kernels calculated at the population level show few, if any, clusters with archetypical opponent profiles. Instead, most colour information seems to be encoded via a combination of differences in spectral gain and kinetics.

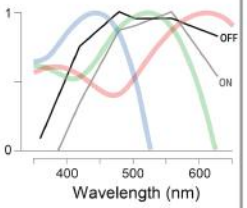


'MESOPIC'

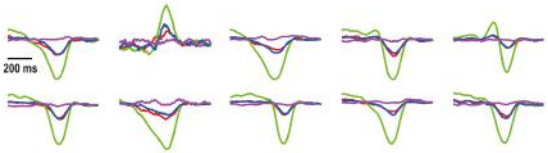


When the intensity of stimuli is reduced 100-fold, rod-mediated responses start to dominate the spectral tuning of both ON and OFF responses at the population level.

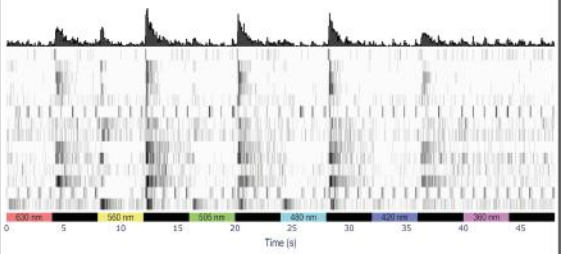
At the same time, the sizeable residual response to long wavelength stimuli is consistent with the high sensitivity of frogs' cones.



Colour kernels also become green-dominated, with little competition from almost perfectly matched red and blue input.

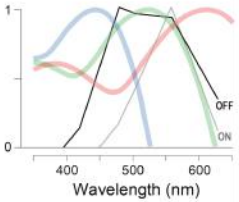


'SCOTOPIC'



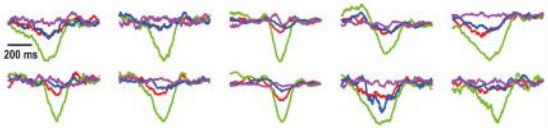
With a further 10-fold reduction in the intensity of stimuli, the number of reliably firing cells decreases drastically, with a visible impact in the quality of the colour kernels.

Interestingly, at this point the retina becomes almost completely Off-dominated to follow the spectral sensitivity of green-rods almost perfectly.



In contrast, On responses appear mostly restricted to "yellow" stimulation.

The spectral tuning of this population On response is approximately consistent with a positive input from green-rod circuits being selectively suppressed at short wavelength by blue-rod circuits.



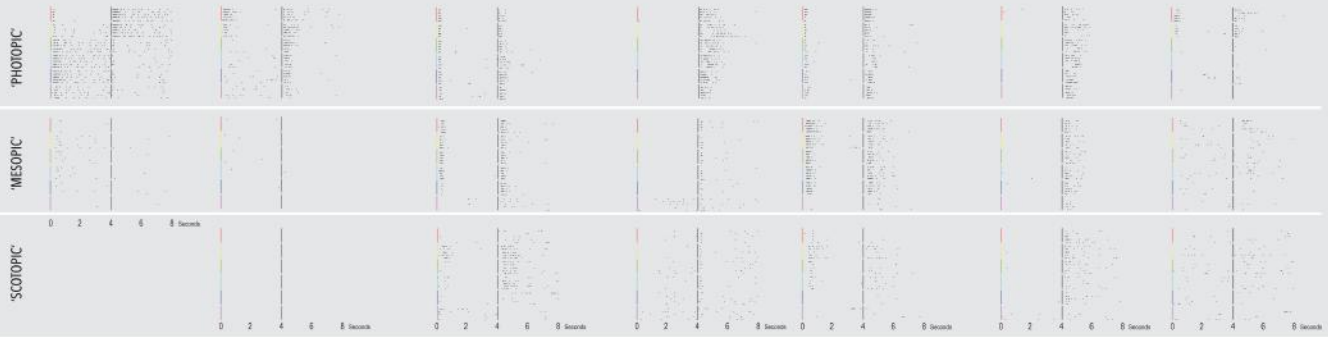
Sneak peek: potential substrates for wavelength comparison at individual cell level under different stimuli intensities

The diversity of wavelength selectivity response patterns in 'typical' opponent cells is quite limited in our sampling so far, and these cells are relatively rare and seem to get lost at low stimulus intensities

However, some other cells seem to use a more subtle strategy for wavelength encoding at low light levels, by selectively suppressing the 'blue-ON' response

The temporal dynamics of the responses can also show differential degrees of wavelength dependence across stimuli intensities, providing a potential additional layer of spectral encoding for the retinal output

Finally, 'dark-proof' spectrally broad cells might act as reference in the context of colour discrimination, whereas narrow-tuned pathways could have the potential to contribute via non-opponent mechanisms, zebrafish-style (see poster 60!)



STAY TUNED!

For upcoming work on:
* 'Blue'-rod circuit anatomy & connectivity (w/ R. Wong lab);
*Imaging of nocturnal, real-life frogs' natural scenes (w/ T. Euler lab)

Opening image modified from © 2003 Alexander Haas

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